

Assignment 3 DCCN (CS221)

Submit on or before 22/09/2026

1. Given the data word 101001111 and the divisor 10111, show the generation of the CRC codeword at the sender site (using binary division). Verify your answer at the receiver side assuming no bit is corrupted.
2. For transmission of data word 1011, find the transmitted codeword using the (7,4) Hamming code with even parity.
3. Discuss the mechanism of stop and wait ARQ. Distinguish between Go Back N ARQ and selective repeat ARQ with examples.
4. Discuss the importance of multiple access. Explain the operations of CSMA/CD and CSMA/CA.
5. Write short notes on (a) IEEE 802.3 standard Ethernet and (b) IEEE 802.11 WLAN.